

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)}$$

$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$

$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & 0 \\ -M_{yzy} & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$

$$M_{yzy} = \mathbb{M}_z^{(3)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxx} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xyz} & -M_{xyz} & 0 & 0 & 0 & -M_{xxx} \\ M_{xxx} & -M_{xxx} & 0 & 0 & 0 & M_{xyz} \\ 0 & 0 & 0 & -M_{xxx} & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{M}_{4a}^{(1)}$$

$$M_{xxx} = \mathbb{M}_{4b}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & 0 & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & 0 & 0 & -M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{T}_{3a}^{(1)}$$

$$M_{xxzx} = -2\mathbb{T}_{3b}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & M_{xxzx} & 0 \\ -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxzx} & -M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{T}_{3a}^{(2)}$$

$$M_{xxzx} = \mathbb{T}_{3b}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3a}^{(3)}$$

$$M_{yzxy} = \mathbb{T}_{3b}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 & M_{xxzy} & M_{xxxz} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 & -M_{xxzy} & -M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & -M_{xxxz} & M_{xxzy} & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} + \mathbb{T}_{3a}^{(2)}$$

$$M_{xxzx} = \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} + \mathbb{T}_{3b}^{(2)}$$

$$M_{xxzy} = \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} - \mathbb{T}_{3a}^{(2)}$$

$$M_{xxxz} = \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} - \mathbb{T}_{3b}^{(2)}$$

$$M_{yzxx} = \mathbb{M}_{4a}^{(1)} - 2\mathbb{T}_{3a}^{(1)}$$

$$M_{yzxy} = -\mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3a}^{(3)}$$

$$M_{yzxy} = \mathbb{T}_{3b}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{yyz} \\ 0 & 0 & 0 \\ -M_{xxz} & M_{yyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyz} \\ M_{xxz} & -M_{yyz} & 0 \\ 0 & 0 & -M_{xxz} \\ -M_{yyz} & -M_{xxz} & 0 \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_{4b}^{(1)}$$

$$M_{yyz} = -\mathbb{M}_{4a}^{(1)}$$